Chapter #9

Why Bitcoin's Intrinsic Value Is More Than Skin Deep

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9.0 Why Bitcoin?

Bitcoin is a game-changer in the financial world, particularly in parts of the world where the traditional banking system is ineffective. In poor communities, traditional banks are often unwilling to cater to the needs of the people due to the high costs of compliance imposed by regulations. As a result, a significant proportion of the population is left without access to essential financial services. Additionally, cross-border remittances into countries like El Salvador are not only expensive but also time-consuming. The fees associated with these transactions and the delay in processing times can be devastating for those who rely on these funds for their daily needs. Moreover, people in unbanked communities are unable to access investments and assets to protect against inflation, further perpetuating their financial insecurity. In light of these issues, ***Bitcoin*** provides a solution that addresses the immediate needs of these communities. It enables the transfer of funds quickly and efficiently, without the need for intermediaries and at a fraction of the cost. Furthermore, it offers a way for people in unbanked communities to store value and protect against inflation.

9.1 The Future of Bitcoin

"**Hyperbitcoinization**" is a theoretical future where ***Bitcoin*** becomes the dominant global currency. This would mean that bitcoin would be used by everyone, everywhere, and for everything - from buying coffee to paying bills and even to buy a house.

The growing interest in ***Bitcoin*** by billionaires, countries, and governments highlights the potential impact of its widespread adoption on the economy and society. Here are some of the benefits of a hyper-bitcoinized world:

1. A Revolution in the Remittance Market: The remittance market involves the transfer of funds from one party to another, often across international borders. Despite declining costs, remittances remain relatively expensive compared to domestic bank transfers, especially for smaller amounts. ***Bitcoin*** has the potential to revolutionize the remittance market by reducing costs to near-zero through its Lightning Network layer 2 protocol. The Lightning Network offers fast and low-cost transactions, making it well-suited for the remittance market and addressing the high costs and other challenges associated with remittances, such as slow settlement times and restrictions on business hours.

2. A Self-Sovereign Future: A self-sovereign future is one where individuals have full control over their own digital identity and assets. It could lead to greater financial inclusion, privacy, and security, and increase the value placed on privacy in transactions.

3. Changes in Monetary Policy: If ***Bitcoin*** were to become widely adopted, it could challenge the ability of governments to control the money supply through traditional monetary policy tools, leading to changes in monetary policy management and implementation. It could also increase financial inclusion, equality, and opportunities, as well as reduce the ability of governments and financial institutions to manipulate the economy.

4. A Reliable Store of Value: ***Bitcoin***'s digital scarcity makes it a reliable store of value, which could encourage more people to use it as a means of saving for the future.

5. Enhanced Transparency and Traceability: The tamper-proof and immutable record of all transactions on the blockchain could increase transparency and accountability in various industries and sectors.

6. Improved Cybersecurity: The decentralized structure of Bitcoin makes it less vulnerable to hacking and data breaches, improving overall security.

7. Reducing Carbon Footprint and Promoting Renewable Energy: By making the process of mining for bitcoin more sustainable and environmentally friendly, miners can help to reduce its carbon footprint and promote the use of renewable energy sources. This aligns with important environmental, social, and governance (ESG) considerations.

9.1.1 The Lindy Effect

The Lindy Effect is a simple theory that says the longer something has been around, the more likely it will continue to be around in the future. This theory can be applied to many things, including ***Bitcoin***.

Bitcoin, a decentralized digital currency that has been around since 2009, is a prime example of the Lindy Effect in action. Despite facing numerous challenges over the years, including technological changes, security breaches, and government regulations, Bitcoin has continued to grow in popularity and has been adopted by an increasing number of businesses as a means of payment.

Acing the Test of Time

One of the key reasons for ***Bitcoin***'s longevity and continued use is its decentralized nature. This means that it operates as a secure and transparent financial system without the need for intermediaries, making it appealing to individuals who value financial privacy and control. In addition, ***Bitcoin***'s ability to operate as a secure store of value has also contributed to its growing popularity and acceptance.

Another factor that contributes to ***Bitcoin***'s longevity is its resistance to change and competition. Changes to the network consensus rules require the majority of users to agree to the update, making it difficult to achieve consensus and leading to only updates that the overwhelming majority of network participants agree upon being implemented. Additionally, despite the existence of many competing cryptocurrencies, none have yet been able to match ***Bitcoin***'s longevity or achieve the same level of network effects.

Bitcoin's hash rate has been increasing exponentially over the years and the distribution of mining is also becoming more widespread. The number of users joining the ***Bitcoin Network*** has also increased at an exponential rate, with an estimated 140-190 million users now a part of it. These factors, combined with its continued popularity and usefulness, suggest that ***Bitcoin*** is likely to continue being used and trusted into the future.

9.2 Using Bitcoin for More Than Just Digital Money

Bitcoin has gained popularity for various reasons beyond just being a means of making money. Some users are driven by the idea of creating a financial system that is free from central control, while others simply want to benefit financially.

Bitcoin also allows for the creation of unique **digital artifacts** known as **Satoshi inscriptions**. These inscriptions, which can include text, images, videos, audio, and software, are stored on the ***Bitcoin*** *blockchain*, making them immutable, secure, and decentralized. The unique identification of each Satoshi is made possible through **Ordinals**. Unlike traditional NFTs, these inscriptions don't require a separate infrastructure or token, which further enhances their security and decentralization.

The combination of ***Bitcoin*** and AI can be utilized for various applications such as cryptocurrency trading, security, and market analysis.

Bitcoin's lightning network has made faster and more secure financial payments possible. For example, atomic swaps enable people to exchange one cryptocurrency for another without the need for an intermediary. RSK, a platform built on top of the ***Bitcoin*** *blockchain*, also allows for the creation of smart contracts and decentralized apps, which opens up new possibilities for what can be built on top of ***Bitcoin***.

As these technologies continue to be developed and improved, exciting things are expected to come in the future.

9.3 The Challenges

Bitcoin Core is a powerful and widely-used implementation of the ***Bitcoin*** protocol. However, there are a few areas in which it could be improved:

1. Scalability: As the number of users and transactions on the network grows, the amount of data that needs to be stored and processed by nodes can become quite large. This can slow down the validation of transactions and make it more difficult for new users to join the network.

2. Privacy: While ***bitcoin*** transactions are pseudonymous, the blockchain is publicly accessible, which means that it is possible for third parties to track the flow of funds and identify users. There are some proposed solutions to this problem, such as the use of **coin mixing** and stealth addresses, but they are not yet widely adopted.

3. Usability: For the average user, the process of setting up and using a full node can be quite technical and daunting. Simplifying the user experience and making it more accessible to a wider range of people could help to increase adoption.

4. Decentralization: ***Bitcoin***'s current consensus algorithm is **Proof-of-Work**, which can be mined by specialized and large mining farms. This can lead to a concentration of mining power and threat to the decentralization and security of the system.

5. Security: While ***Bitcoin Core*** is open-source, which means that its code can be audited by anyone, it's still possible for bugs or vulnerabilities to be introduced into the code. Continuously auditing and improving the security of the software can help to protect users from potential attacks. For example, if an attacker were to generate a private key that corresponds to a large number of bitcoin, they could steal those bitcoins.

Overall, while ***Bitcoin Core*** is a solid piece of software, ongoing development and research is essential to address these areas of improvement and ensure that the network remains secure, decentralized and widely adopted.

9.3.1 The Regulatory Environment for Bitcoin

The cryptocurrency market has faced numerous challenges in recent years, including the collapse of FTX in 2022 and the fall of stablecoins TerraUSD and LUNA earlier in the same year, leading to significant losses and a decline in investor confidence. The risks associated with investing in cryptocurrencies include volatility, difficulty in evaluating assets, custodial risks, unregistered assets and providers operating outside of regulatory frameworks, and unpredictable regulations.

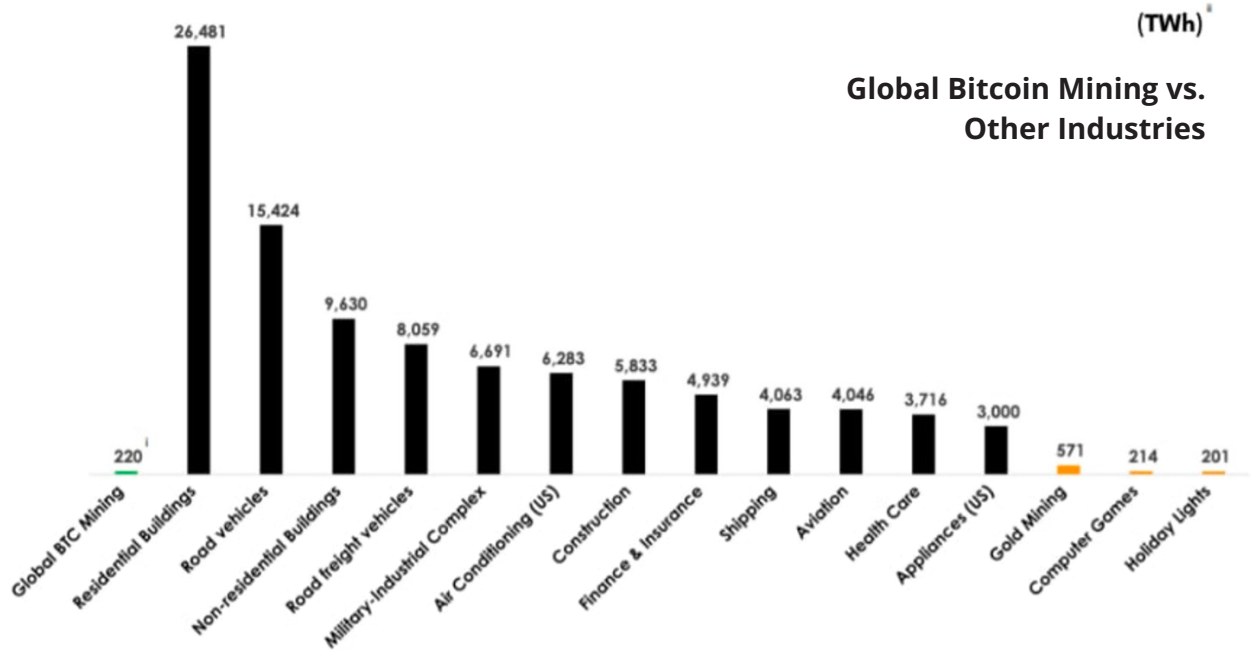
The regulation of cryptocurrencies, including ***Bitcoin***, has been a topic of debate among governments and financial regulators worldwide. While some have banned cryptocurrencies, others have sought to regulate them in a way that balances innovation and consumer protection. The U.S. SEC Chairman

Gary Gensler recently stated that the regulation of the cryptocurrency market is getting closer and that ***Bitcoin*** will be considered a commodity. According to the SEC, many tokens in the market have the key attributes of securities and will fall under the jurisdiction of the SEC, while ***Bitcoin*** falls under the oversight of the Commodity Futures Trading Commission (CFTC) as a commodity. The SEC has work to do to introduce comprehensive laws that protect investors, and this decision by the SEC chair is seen as positive by some investors, leading to the expectation of gradually rising prices. Despite the current market volatility, some investors view this as a buying opportunity and believe in the future of digital currencies as a borderless, decentralized, tamper-proof, and unconfiscatable form of money.

9.3.2 Understanding the Energy Usage of Bitcoin Mining

Bitcoin mining uses a lot of energy, around 79 terawatts-hours per year. However, this does not necessarily mean that it is a waste of energy or harmful to the environment. ***Bitcoin*** mining can help to utilize unused capacity of energy, especially in remote or inaccessible places. Additionally, most Bitcoin mining is done with renewable energy such as hydroelectric, solar, wind, and geothermal. This helps to make the production and research of these energy sources more profitable. Furthermore, Bitcoin mining provides security for the Bitcoin network, enabling people to have access to secure and accessible money.

However, it's important to note that the energy usage is determined by the competition among miners, not the number of transactions. The digital signature validation process, which is a small part of mining, uses minimal energy. The energy usage of ***Bitcoin*** mining is high, but it is not as high as other industries like the traditional financial system or gold mining and recycling. Miners are also increasingly using clean and renewable energy sources like geothermal and hydroelectric power to power their mining operations.



The key to reducing the environmental impact is to drive the demand for green energy, and as the industry grows, it's leading to innovation in clean energy production and decreased pollution. It's also important to note that the source of energy used by miners greatly impacts the ecological impact. As the technology and industry evolve, more miners are using renewable energy sources such as hydroelectric, solar, and wind power, which greatly reduces the environmental impact.

9.4 The Risks

Bitcoin can offer great freedom, but it's important to remember that with great power comes great responsibility. There are risks involved in using ***bitcoin***, so it's essential to understand these risks and take proactive steps to protect your funds.

1. Volatility: The value of ***bitcoin*** can be highly volatile and can change a lot in a short period of time, which can lead to significant losses for investors.

2. Lack of regulation: ***Bitcoin*** is not regulated by governments or financial institutions, which means there is little oversight to protect consumers.

3. Security risks: Bitcoin exchanges and wallets can be subject to hacking and theft, which can result in the loss of funds for users.

4. Scams: There are many scams related to ***Bitcoin*** that can lead to the loss of funds for investors.

5. Illicit activities: ***Bitcoin*** has been used for illegal activities like money laundering and buying illegal goods on the dark web.

6. Lack of understanding: ***Bitcoin*** is complex and can be difficult to understand for the average person, which can lead to poor decision making and potential losses.

7. Lack of acceptance: ***Bitcoin*** is not widely accepted as a means of payment, which limits its usefulness in everyday life.

8. Technical risks: ***Bitcoin*** is subject to technical risks like bugs and errors, which could lead to problems and potentially a loss of value.

9. Quantum computing: Quantum computing could potentially compromise the security of ***Bitcoin*** by breaking the encryption used to secure transactions and wallets.

Quantum computing is a way of doing computer calculations that's different from the way most computers work today. Instead of using just "on" and "off" states like traditional computers, quantum computers use "**qubits**" that can be in many states at the same time. This makes quantum computers potentially much faster at certain types of calculations than regular computers.

10. Digital Threats: Hackers can exploit your internet connection to access your **private keys** and sensitive data, which includes hacking software wallets, clicking malicious links, and falling for spyware scams.

11. Social Engineering Scams: Scammers can manipulate you into confirming transactions by posing as customer service agents or creating a false sense of trust, so it's important to be cautious and not share your recovery phrase.

12. Blind Signing: Lack of transparency can lead to blind signing, where you agree to transactions without fully understanding the details, so it's important to educate yourself on the latest scams and choose a wallet that displays full transaction details.

13. A **51% attack** is a potential security threat to the ***Bitcoin Network***, and it occurs when a single miner or group of miners control more than 50% of the total computational power or hashrate of the network. This enables them to take control of the network and potentially manipulate the blockchain by either preventing new transactions from being added or modifying transactions in their favor.

If an attacker were to successfully perform a 51% attack, they could double-spend ***bitcoin***, which means they could spend the same bitcoin more than once. This would allow them to effectively steal bitcoins or commit fraud on the network. However, executing a 51% attack is incredibly difficult and costly, as it would require controlling a significant amount of computational power, and the costs involved in obtaining such power could outweigh any potential gains from the attack.

It's important to note that the ***Bitcoin Network*** has never been successfully attacked in this way, but the possibility of a 51% attack is always present, and it highlights the importance of ensuring that the network remains decentralized and secure.

To protect your ***Bitcoin*** security, use an offline wallet, read full transaction details, and continuously educate yourself on the latest threats. Don't let ignorance and a false sense of trust compromise your hard-earned assets.

While the risk of quantum computing to ***Bitcoin***'s security is real, it's important to remember that it's still a speculative threat and it's uncertain when or if it will become a reality. A 51% attack on ***Bitcoin*** is a concern, but it would be costly and not very beneficial for the attacker. More efficient and cost-effective methods of attack, such as DDoS, would be more likely for a rational actor seeking to commit fraud.

9.5 Trading and Investing in bitcoin

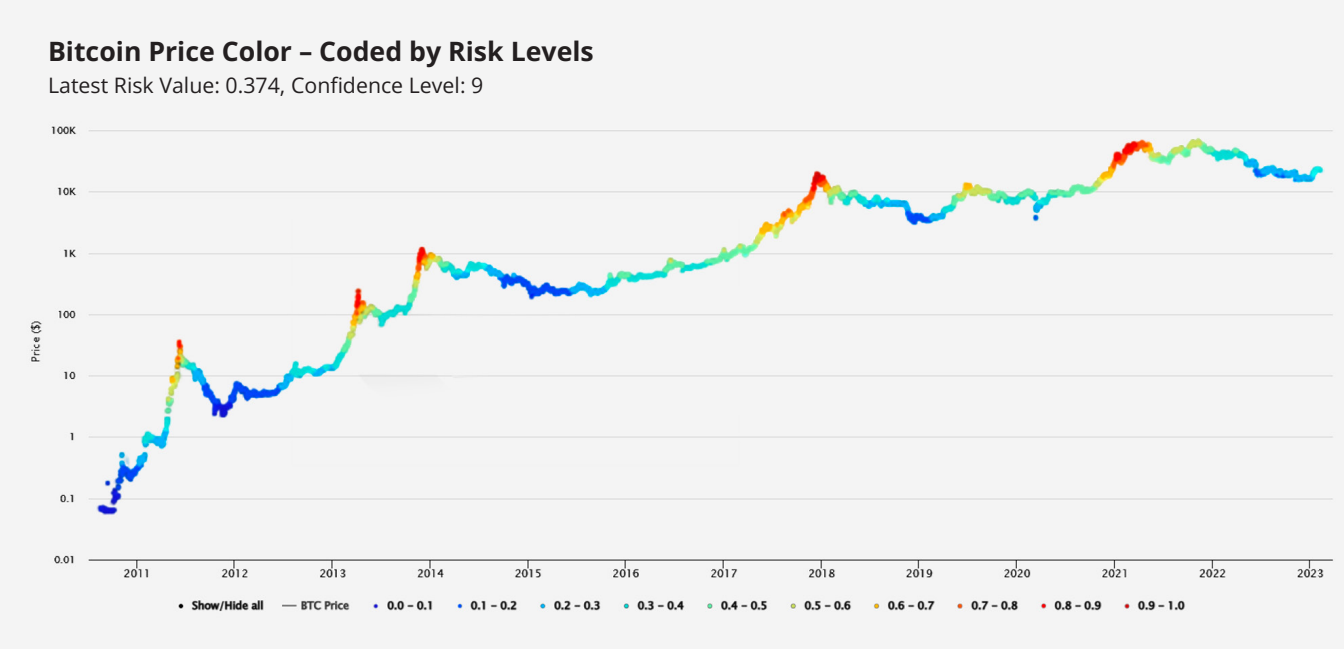
When it comes to investing in cryptocurrency, ***bitcoin*** is the safe and reliable choice, and the one that aligns with the values and principles of a decentralized future.

Market trends refer to the general direction that the market is moving in. A bullish trend is when the market is on an upward trajectory, while a bearish trend is when the market is on a downward trajectory. This is usually associated with investor optimism and an expectation that prices will continue to rise. In contrast, a bearish trend is when the market is on a downward trajectory, characterized by lower highs and lower lows. This is usually associated with investor pessimism and an expectation that prices will continue to fall.

Technical analysis is not a perfect science, and past performance is not always indicative of future results. It should be used in conjunction with other forms of analysis, such as fundamental analysis and market sentiment, to make informed trading and investment decisions.

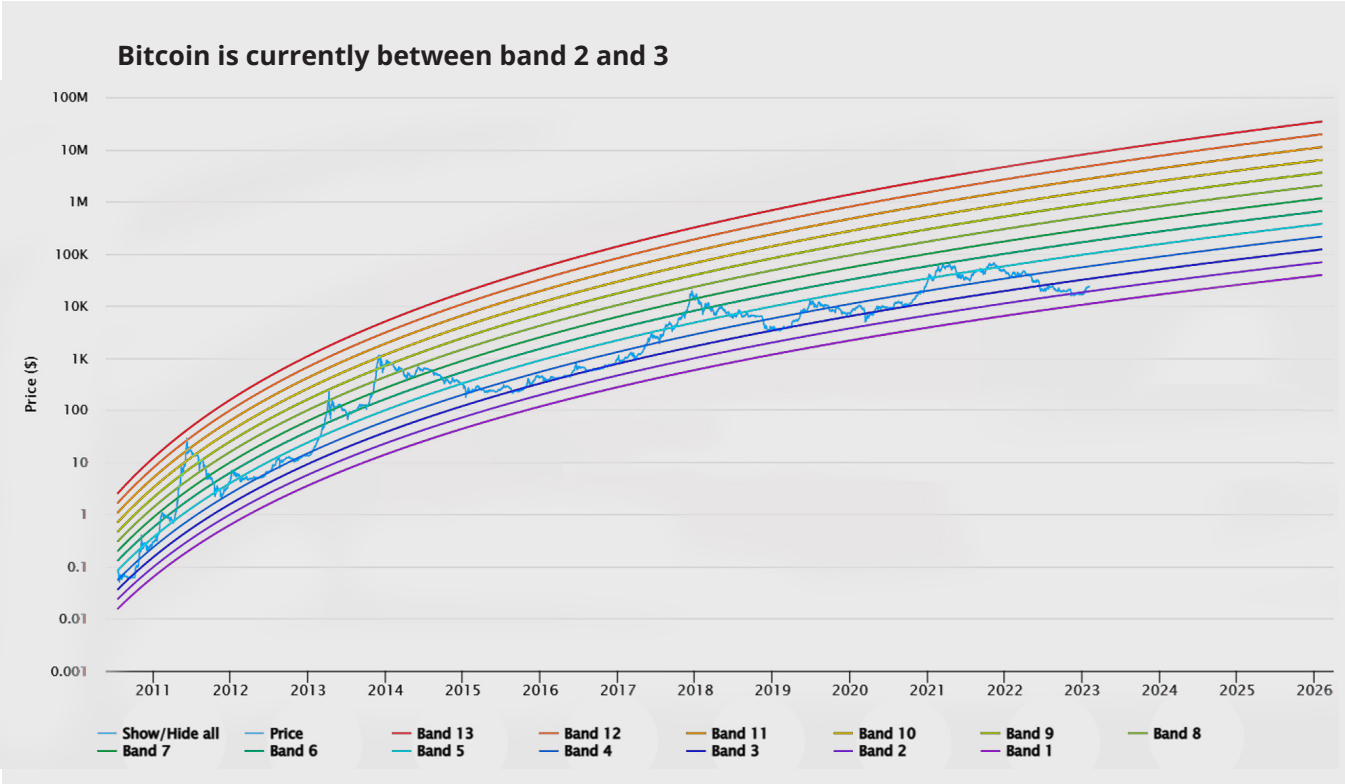
The Risk Metric chart, created by Benjamin Cohen, is a quick and intuitive way to understand market sentiment and asses potential buying or selling opportunities for bitcoin. This chart displays the price of the assets and assigns a color-coded value to represent the risk associated with that price. The risk values range from 0 to 1, with darker red colors indicating higher risk and darker blue colors indicating lower risk.

The purpose of the **Risk Metric** is not to predict market tops or bottoms, but rather to identify areas that may be attractive for buying or selling in the long term. A low risk score suggests that bitcoin may be undervalued and may present an opportunity to buy, while a high risk score suggests that it may be overvalued and may present an opportunity to sell.



The logarithmic market price is a method of visualizing the price movements of an asset, such as bitcoin, over time. This approach uses a logarithmic scale on the y-axis to better reflect the exponential growth that is often seen in asset prices.

The logarithmic market price is being used to track the price movements of bitcoin over time and to identify potential peaks and accumulation zones. The market cycles referred to in the example are periods of price increase and decrease, and the rainbow bands are used to illustrate the relative magnitude of these price movements.



The logarithmic market price can be useful for identifying potential accumulation zones, or periods where the price may be relatively low and provide a good opportunity for buying. In the example, the zones between band 3 and 4 are identified as good accumulation periods for market cycles 3 and 4.

Market cycles in ***Bitcoin*** refer to the recurring pattern of growth and contraction in its price and market activity. It is characterized by periods of speculation and hype, followed by correction and consolidation. Some analysts argue that the cycles are strongly correlated to the halving events.

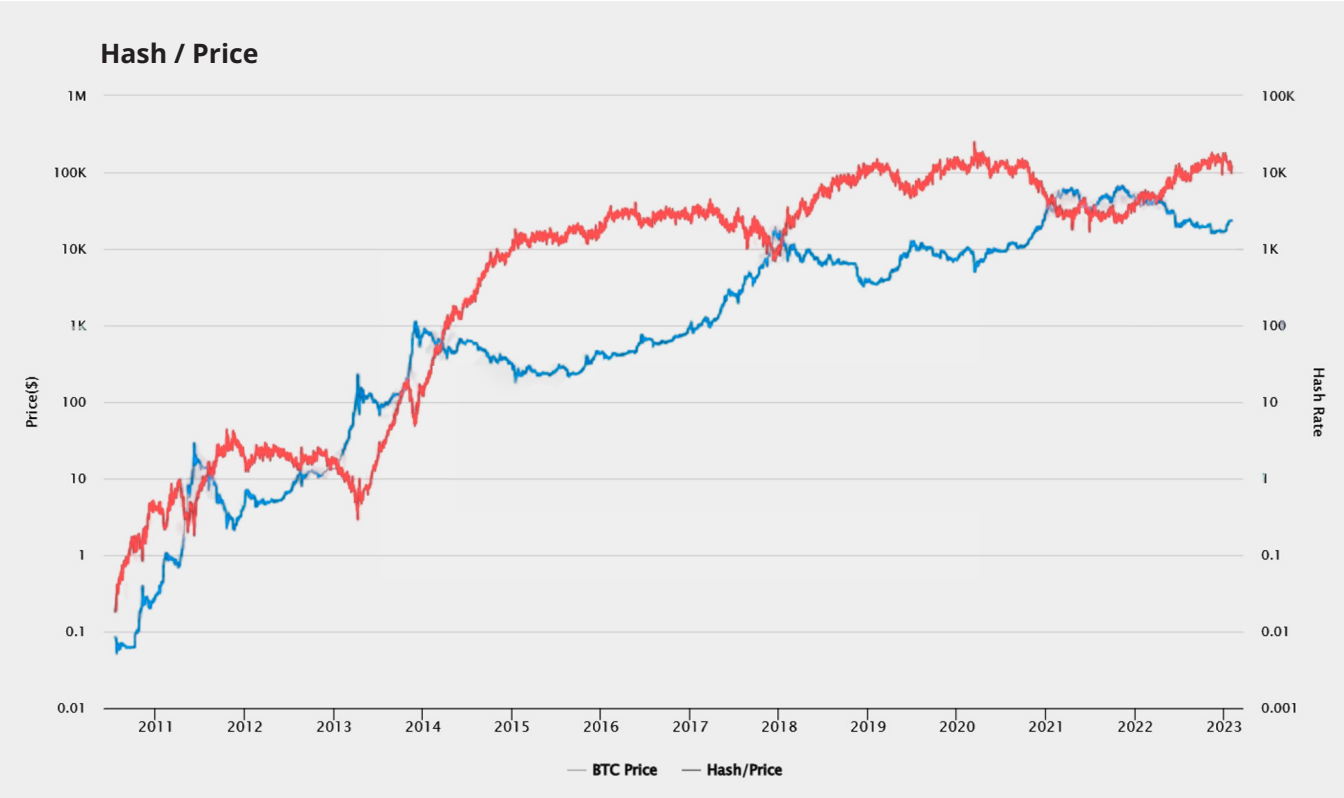
It is important to note that while the logarithmic market price can provide valuable insights, it is only one of many tools that can be used to analyze market trends and price movements, and should be used in conjunction with other analysis methods to form a more complete understanding of the market. Additionally, market conditions are constantly changing, and past performance is not a guarantee of future results.

The Hash/Price ratio and Price/Hash ratio are metrics used to compare the growth of the ***Bitcoin*** price and the growth of the ***Bitcoin Network***'s computational power, or hash rate. These metrics are used to help understand the relationship between the two, and how changes in one may affect the other.

When the price of ***bitcoin*** increases at a faster rate than the hash rate, the Hash/Price ratio decreases and the Price/Hash ratio increases. This means that the price of ***bitcoin*** is growing faster than the computational power of the network, which could indicate increased demand for ***bitcoin***.

However, near local peaks, when the price of ***bitcoin*** is increasing quickly, you may see sudden drops in the Hash/Price ratio. This is because the growth in price outpaces the growth in computational power, leading to a decrease in the Hash/Price ratio.

On the other hand, if both the hash rate and price of ***bitcoin*** decrease or increase at the same relative rates, the ratios will stay constant. This means that the computational power of the network and the price of ***Bitcoin*** are growing at the same rate.

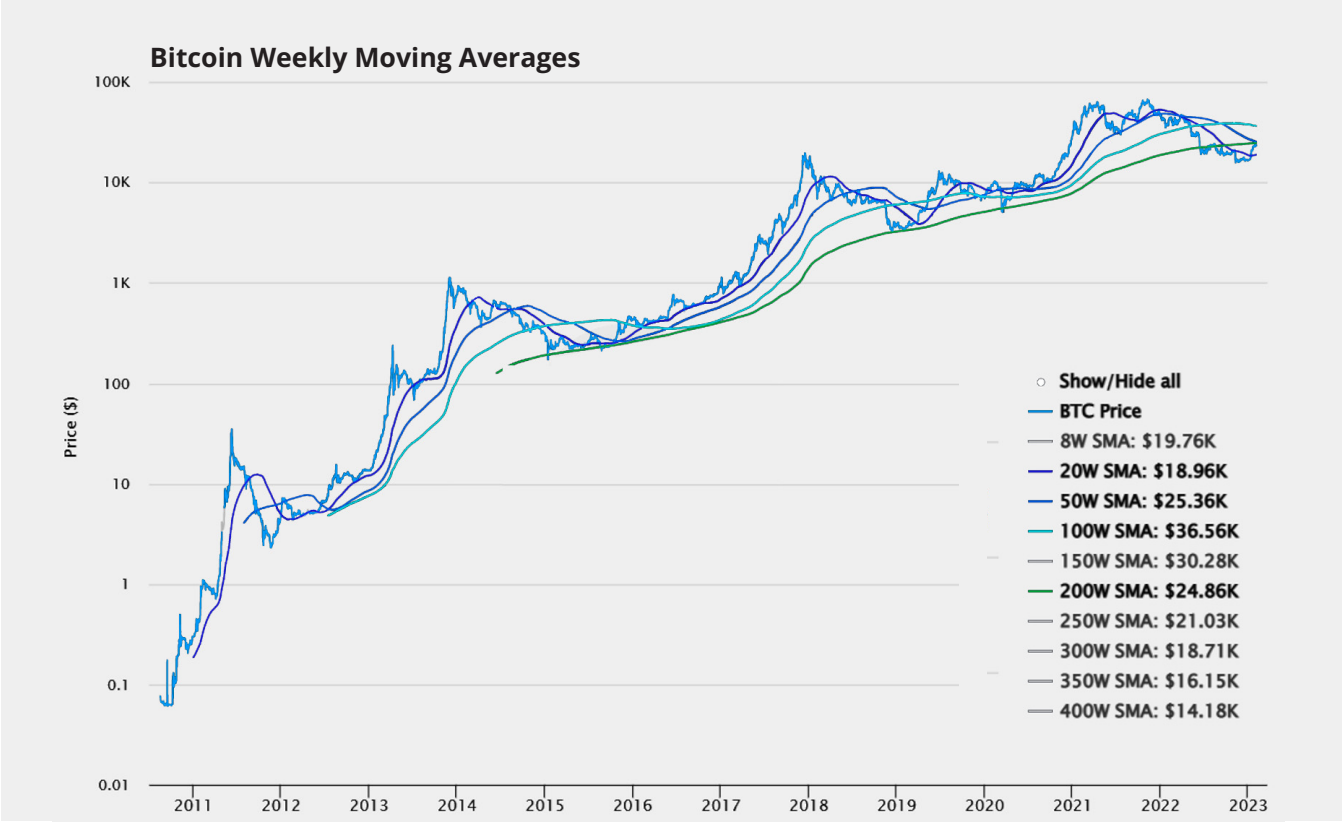


If the hash rate of the ***Bitcoin Network*** is increasing at a faster rate than the price of ***bitcoin***, the Hash/Price ratio will increase and the Price/Hash ratio will decrease. This could indicate that the network is becoming more secure and more capable of processing transactions, which could have a positive impact on the price of ***bitcoin*** in the future.

Trend lines are used to identify a current trend in the market. They are formed by connecting two or more price points and are used to indicate a level of support or resistance. A trend line that is angled up is considered bullish, while a trend line that is angled down is considered bearish.

Moving averages are used to smooth out the volatility of a security's price over a specific period of time. They are calculated by adding up the closing prices of a security over a specific number of periods and then dividing by the number of periods. A moving average can be used to identify the direction of a trend and can also be used to generate buy and sell signals. **Dollar-cost averaging** (DCAing) below shorter term moving averages like the 100-week and 50-week SMA can provide more entry points, but it may leave you at a loss in the short term.

Dollar-cost averaging (DCA) is a strategy of investing a fixed amount of money into a particular asset at regular intervals, regardless of the price.



The goal of DCA is to reduce the impact of market volatility on an investment portfolio by spreading out the purchases over time, instead of buying all at once.

- For example, an investor may decide to invest $100 into a cryptocurrency asset every month. If the price of the asset is high, the investor will purchase fewer units, and if the price is low, the investor will purchase more units. Over time, this approach can lead to a lower average cost per unit of the asset, and thus reduce the impact of short-term price fluctuations.

DCA can be used in a variety of investments, including stocks, bonds, and commodities, and is often recommended for individuals who are just starting to invest and want to minimize the risk of market volatility.

It's important to note that DCA does not guarantee a profit or protect against loss in a declining market, and should be combined with thorough research and market analysis. Additionally, investors should consider their own financial goals and risk tolerance when deciding on the best investment strategy.

Indicators such as the **RSI** and the **MACD** are used to identify overbought and oversold conditions and potential trend changes. The RSI compares the magnitude of recent gains to recent losses to determine overbought and oversold conditions. The MACD is calculated by subtracting the 26-period exponential moving average (EMA) from the 12-period EMA, and then plotting a 9-day EMA of the result. It is used to identify changes in momentum and trend.

It is important to note that these metrics are only one of many tools that can be used to analyze market trends and price movements, and should be used in conjunction with other analysis methods to form a more complete understanding of the market. Additionally, market conditions are constantly changing, and past performance is not a guarantee of future results.